

Maenrat Phaiphon

แมนรัตน์ พายผล

Computer Science Graduate | Nonthaburi, Thailand



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CS graduate with experience in machine learning, backend development, and data-driven applications. Skilled in building ML models, APIs, and dashboards. Fast learner, adaptable team player, seeking an entry-level AI, Data or Fullstack role.

Skills & Proficiencies

Computer Languages: C, Java, Python, SQL, Dart (Flutter) , HTML and CSS, JavaScript(Vue js)

AI: Tensorflow, Pytorch, RAG, LLM APIs (OpenAI, Groq, Gemini), Vector Database, Prompt Engineering

Data Tools: Cloud (AWS and CloudFlare), Kafka, Spark, Airflow, Lookers studio, DBT

Tools: FastAPI, Docker, Git/Github, Linux, Postman, AI tools (Claude / opencode), Streamlit

Languages: Thai (Native) , English (B2 CEFR levels by STEP)

Work Experience

Data Scientist - Medihealth Act (Health tech startup) | Contract

Nov 2025 - Feb 2026

- Designed and developed an AI-powered workforce scheduling backend using Google OR-Tools, implementing constraint-based optimization (availability, coverage, and workload balancing) that reduced schedule generation time from hours to approximately 5 minutes. The solution was deployed for real-world use at a hospital in Bangkok
- Built Looker Studio dashboards using SQL to analyze user behavior and support business decisions
- Conducted manual system testing and User Acceptance Testing (UAT) to identify bugs, edge cases, and system failures before release

Deep Learning Researcher| Silpakorn University (IEEE ICSEC 2025)

May 2025 - Nov 2025

- Developed and benchmarked 27 deep learning classifiers across CNN, EfficientNet-B0, and ResNet50 architectures using TensorFlow and PyTorch for AI-generated image detection
- Designed and analyzed experiments comparing multiple image preprocessing strategies, identifying the best-performing model through accuracy, precision, recall, and F1-score evaluation
- Built a FastAPI + Gradio inference service that integrated multiple classifiers into a unified prediction pipeline for real-time image classification
- Co-authored an IEEE ICSEC 2025 publication, contributing to model development, experimentation, and an ensemble pipeline achieving 97% classification accuracy

Project

Agentic Loan Applicant Screening System

- Designed and developed a 6-agent loan screening system using LangGraph, implementing short-circuit routing to reduce unnecessary LLM calls by 66% while classifying applicants into four decision outcomes
- Implemented a three-layer AI safety framework combining prompt guardrails, deterministic business-rule validation, and mandatory human review to improve the reliability of LLM-generated recommendations.
- Developed an asynchronous FastAPI backend with job-queue polling, audit logging, and officer feedback tracking, following compliance-by-design principles with no automatic loan approvals

AI Data Copilot | Rag system project

- Built an AI-powered copilot that **converts natural language queries into SQL** and retrieves insights from structured databases and knowledge bases
- Designed a natural language → SQL pipeline using LLMs, enabling non-technical users to query data without writing SQL
- Designed the system with read-only queries, input validation, and data masking to prevent unsafe operations
- Built a web interface (Vue) to display results and generated SQL queries

Education

2021 - 2025

Silpakorn University | Bachelor of science : Computer Science

- Supported a Java programming course as a lab assistant, helping students understand core concepts and adopt best coding practices
- Selected for Japan Scholarship Program, participated in international academic exchange and collaborative learning